

SOLID Principles & Design Patterns - Quick Revision Notes

QUICK REVISION & EXAM CHEAT SHEET

Subject: Software Engineering | Chapter: Chapter 2: System Architecture, SOLID Principles & Design Patterns | Academic Year: 2026-2027

SOLID Acronym Breakdown

- * S (Single Responsibility): One job per class.
- * O (Open/Closed): Extend behavior via inheritance/interfaces, do not edit tested classes.
- * L (Liskov Substitution): Child class must honor all parent contracts without throwing unexpected exceptions.
- * I (Interface Segregation): Split fat interfaces into fine-grained interfaces.
- * D (Dependency Inversion): Inject dependencies as interfaces (Inversion of Control / DI).

GoF Design Patterns Quick Lookup

- * Singleton: Private constructor + static getInstance(). Used for thread pools, caches, loggers.
- * Factory: Encapsulates object instantiation logic behind a common interface.
- * Adapter: Wrapper enabling incompatible interfaces to collaborate.
- * Observer: Publisher-subscriber pattern used in event-driven systems and UI listeners.
- * Strategy: Injects interchangeable algorithms (e.g., PaymentStrategy: PayPal, Stripe, Crypto).